

The gap I'm trying to fill is that NutriBench evaluates the accuracy of carbohydrate estimation by Large Language Models, but does not assess whether these predictions are well-calibrated in terms of confidence. In other words, while existing work measures how wrong a model is, it does not measure how sure the model is when it is wrong.

My question: When LLMs estimate carbohydrates incorrectly, are they confidently wrong or uncertainly wrong?

Why this matters for Type 1 Diabetes (T1D): Accurate carbohydrate estimation is directly tied to insulin dosing. If a model gives a confident but incorrect estimate, it can lead to insulin miscalculation without any warning, potentially causing dangerous blood glucose levels (hypoglycemia or hyperglycemia). In contrast, an uncertain wrong estimate could signal risk and allow users to double-check. Therefore, understanding and detecting confidence in LLM predictions is critical for safe real-world deployment in healthcare settings.